Question 17: Word Frequency Analysis from CSV

17. Scenario: You are a data analyst working for a marketing research company. Your team has collected a large dataset containing customer feedback from various social media platforms. The dataset consists of thousands of text entries, and your task is to develop a Python program to analyze the frequency distribution of words in this dataset. Your program should be able to perform the following tasks:

 Load the dataset from a CSV file (data.csv) containing a single column named "feedback" with each row representing a customer comment.

 Preprocess the text data by removing punctuation, converting all text to lowercase, and eliminating any stop words (common words like "the," "and," "is," etc. that don't carry significant meaning).

 Calculate the frequency distribution of words in the preprocessed dataset.

 Display the top N most frequent words and their corresponding frequencies, where N is provided as user input.

 Plot a bar graph to visualize the top N most frequent words and their frequencies.

Aim: To preprocess customer feedback, calculate word frequencies, display the top N words, and plot a bar graph.

Algorithm:

1. Load data.csv using Pandas.
2. Convert text to lowercase and remove punctuation.
3. Remove stop words.
4. Calculate word frequencies.
5. Display the top N words and plot a bar graph.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
import re
from collections import Counter
df = pd.read_csv("data.csv")
stop_words = {"the", "and", "is", "a", "an", "in", "to", "of", "it"}
text = " ".join(df["feedback"].dropna()).lower()
text = re.sub(r"[^\w\s]", "", text)
words = [w for w in text.split() if w not in stop_words]
frequency = Counter(words)
n = int(input("Enter N: "))
top_words = frequency.most_common(n)
print("Top Words:")
for word, count in top_words:
    print(word, ":", count)
if top_words:
    words, counts = zip(*top_words)
    plt.bar(words, counts)
    plt.xlabel("Words")
    plt.ylabel("Frequency")
    plt.title("Top N Frequent Words")
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.show()
```

Input (data.csv):

feedback

Product is good

Good product quality

The product is excellent

Sample Output:

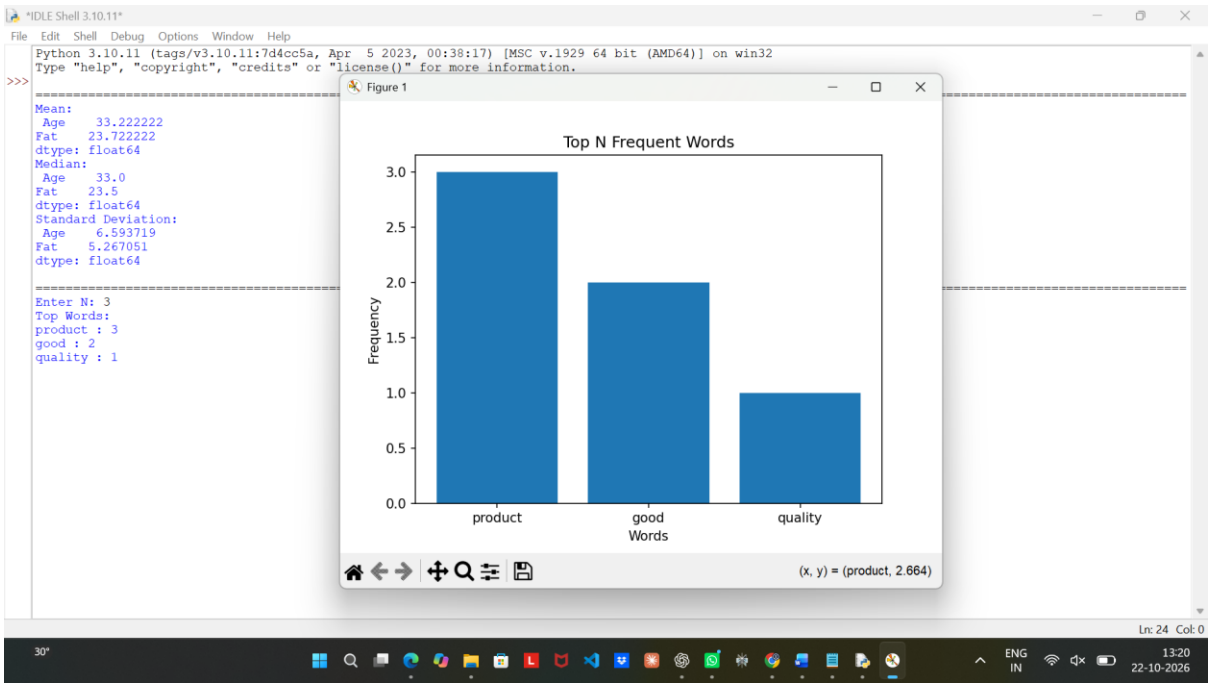
Enter N: 3

Top Words:

product : 3

good : 2

quality : 1



Result: Word frequencies were calculated, and the top N words were visualized using a bar graph.